

Predicting the NCAA Men’s Gymnastics National Champion

Alex Grapsas

Department of Computer Science

William & Mary

Williamsburg, VA

angrapsas@wm.edu

Abstract—Gymnastics is a numerical sport. Every complex movement and impressive feat of strength is translated into tenths of a point, producing a rich quantitative record of every competition. As an NCAA gymnast, this made me wonder: can a machine learn the signals and patterns of an NCAA Championship team? We approached this as a binary classification problem (champion or not) using logistic regression. We constructed a dataset spanning 13 competitive seasons from an undocumented API and engineered domain-specific features informed by gymnastics expertise, including normalization to account for an ever-changing scoring system. Team strategy and the sport’s unpredictable nature make this challenge particularly difficult. Despite this, across most model iterations we correctly identified the 2026 NCAA Champion before the meet occurred, with win probabilities for the top-4 contenders aligning with their final finishing order. The model also correctly predicted the champion in both held-out test seasons. We further examined how early in the season reliable predictions emerge, with conference championship data proving critical. When analyzing the model’s learned feature weights, we uncovered a remarkable statistic that correctly identified the champion in advance for every season examined. It’s the only single indicator of its kind, to our knowledge.

Index Terms—binary classification, logistic regression, sports analytics, feature engineering, era normalization, NCAA gymnastics

I. INTRODUCTION

NCAA Men’s Gymnastics is a small sport. Roughly fifteen programs compete in a single national championship each year, drawn from both Division I and Division III with no distinction made between divisions at the championship level. Top recruits tend to congregate at a small number of historically dominant programs, and only a handful of teams in any given season are realistic title contenders. On the surface, that should make predicting the winner straightforward: pick the team that has been winning all year. The data does not cooperate. In 6 of the 13 seasons from 2013 to 2026, the top ranked team entering the championship did not win it.

Several factors contribute to this difficulty. The first is that elite programs have more flexibility in how they manage their lineups during the regular season than lower-ranked programs do. Teams that are not in danger of missing the championship can choose to rest athletes, particularly top all-arounders who simultaneously compete at elite national and international competitions during the NCAA season. The result is that a championship contender may at times look weaker on paper

than they actually are. This does not mean that top teams are visibly underperforming all season; programs like Stanford and Michigan often still rank near the top of the standings while resting athletes. But the strategic flexibility introduces noise into how regular season scores reflect a team’s true ceiling.

A second source of difficulty is that the scoring system itself shifts every year. Element values, deduction tables, and lineup rules are revised regularly, driven by both international and NCAA level rule changes. The most dramatic recent change came in 2024, when the NCAA reduced counting athletes per event from five to four and FIG (the international governing body) reduced counting skills in routines from ten to eight, a change the NCAA subsequently adopted. Team scores dropped by roughly 80 to 100 points overnight. Smaller adjustments occur every season. Comparing raw scores across years is therefore meaningless, and any model that treats them as comparable will fail.

The motivation for this project is partly personal: as an NCAA gymnast, I follow this sport closely and have a stake in understanding what distinguishes championship teams. The project also serves as a case study in applying machine learning under realistic constraints: a small, severely imbalanced dataset, frequent rule changes that disrupt cross-year comparisons, and outcomes shaped by strategic decisions not directly observable in the data. The methodological choices required to navigate these constraints generalize beyond gymnastics. For the gymnastics community, any patterns this work surfaces may offer one more lens through which to think about what makes a team a contender.

This project asks whether regular season performance data, properly engineered with domain knowledge, can reliably predict the NCAA Men’s Gymnastics National Champion. A secondary question is how early in the season such a prediction becomes reliable. We use logistic regression as the primary classifier and compare its results to a Support Vector Machine and to several naive baselines.

II. APPLIED APPROACHES

A. Data Collection

Data was sourced from Road to Nationals [1], the website that tracks NCAA gymnastics results throughout each season. The site provides no public API documentation. Endpoints

were reverse-engineered by inspecting the minified JavaScript bundles served by the site’s React front-end. Twelve endpoints were identified and used to retrieve team dashboard data, apparatus-level consistency scores, and final meet results for every season from 2013 through 2026, excluding 2020 due to the COVID-19 cancellation.

The first significant challenge was that the same API serves both NCAA and GymACT (a separate collegiate gymnastics organization) teams, with no flag distinguishing the two. Several programs that left NCAA for GymACT, including Arizona State, Iowa, Minnesota, and Temple, returned corrupted consistency data in their post-NCAA seasons, with deltas of 60 to 75 points between raw apparatus sums and reported team totals. We filtered teams by conference identifier, retaining only those affiliated with the Big Ten (`id=1`), ECAC (`id=2`), and MPSF (`id=4`). These three conferences contain only NCAA-affiliated programs, so the filter served as a proxy for current NCAA membership.

While building the dataset we noticed an anomaly: at one Illinois meet in January 2025, the sum of the team’s per-apparatus scores did not match the official team total by a 2.0-point margin. To check whether this was an isolated case, we audited all 1,415 meet-scores in the dataset, comparing apparatus sums against final team scores. Twenty-two rows (1.6%) showed non-zero deltas. Twenty-one were legitimate team neutral deductions (administrative penalties for things like athlete substitutions or equipment violations). The Illinois case remained unique and unexplained, but every other discrepancy was accounted for. We retained the Illinois data point.

The final dataset spans 12 seasons of labeled data (2013–2025, excluding 2020), containing 185 team-season observations and 12 winning team-seasons (one per season). Teams per season ranged from 13 to 17. The 2026 season was held out as a separate prediction set.

B. Feature Engineering

NCAA team scores are the sum of six apparatus scores, which are the sum of the routines in that team’s lineup. Per-apparatus scores were obtained from the `teamconsistency` endpoint and final team totals from the `finalresults` endpoint. We used `teamconsistency` for feature construction and `finalresults` for the champion labels.

The feature set evolved as we worked through the project. Our initial set was derived from straightforward aggregates of regular season meet data: per-apparatus mean, standard deviation, maximum, and slope over the final few meets, as well as season-level analogues of these. We later added features that more closely mirror the NCAA’s own ranking methodology, in particular the National Qualifying Average. Those additions are described below alongside the original features they extend.

Apparatus maximum. The single highest score posted on each apparatus at any meet during the team’s regular season

(6 features). This represents the team’s demonstrated ceiling on each event.

Apparatus standard deviation. The standard deviation of each team’s per-apparatus scores across their valid meets. Standard deviation measures how scattered a team’s scores are: a team with consistent scores across meets has a low standard deviation, while a team with both high highs and low lows has a higher one. We initially included this feature for all six apparatus. After early model iterations, we removed `std_vt` (vault standard deviation) because it was being weighted heavily in ways that produced rankings inconsistent with the eye test, suggesting it was contributing noise rather than signal. We did not investigate the underlying cause and kept the standard deviation feature on the other five apparatus (5 features total). One possibility is that vault variance reflects strategic lineup changes, but more likely the team is simply scoring differently on vault from meet to meet for reasons we could not verify without lineup-level data.

Apparatus score trajectory. The slope of each team’s per-apparatus scores over their final 3 to 4 meets, fit linearly (6 features). This loosely captures the direction a team’s scores are moving at that point in the season.

Season aggregates. Season-level analogues of the above: total mean, total maximum, total slope, and number of valid meets (4 features). Number of meets is included as a feature, but we make no claim about its relationship to team strength; some top teams compete at fewer meets than weaker programs, and the model’s learned weight on this feature was small.

Conference championship total. The team’s total score at their conference championship meet (1 feature). The conference championship is the final regular-season-eligible meet and is treated by every program as the most important meet of the season prior to the NCAA Championship, so it gets its own feature.

NQA per apparatus. The NCAA ranks teams using a National Qualifying Average: drop the highest regular season score on the apparatus, average the next three, then average that result with the conference championship score on the apparatus. We implemented this on each of the six apparatus separately (6 features). NQA is only computable after the conference championship has been competed, since it requires the conference score to exist. For our week-by-week analysis (Section III-F), which uses cutoffs throughout the regular season, we used a simpler 4-score average of the team’s top apparatus scores instead, which also served as the precursor to NQA in the project’s feature engineering. Computing NQA also required identifying which meet was each team’s conference championship for each season; this had to be inferred from meet labels and dates since neither the API nor any single field flagged it directly.

C. Era Normalization

Because the scoring system shifts every season, raw scores are not comparable across years. Rather than try to identify and correct for each individual rule change, we applied within-season z-score normalization to every feature, separately for

each season. This treats the champion as a relative concept: the winning team is the team that posts the highest score at the NCAA Championship final, judged against the rest of the field in that year’s scoring environment. Relative within-season performance is the only signal that generalizes across eras, and applying the same normalization every year handles all rule changes automatically without needing to enumerate them.

D. Model

The task is binary classification: for each team-season, the model predicts whether that team will win the NCAA Championship (the positive class) or not (the negative class). We chose this framing because it directly matches the question we are trying to answer; predicting outcomes other than “champion” was not a goal of the project, although the model’s continuous probability outputs do allow us to compare non-champions on a relative scale.

We used logistic regression as the primary classifier [2]. Logistic regression assigns each team a real-valued score that is a weighted sum of its features, with each feature multiplied by a learned coefficient. That score is passed through the logistic (sigmoid) function, which maps any real number to a decimal between 0 and 1. This output is not a literal win probability; it is the model’s confidence that a team’s feature profile matches the “champion” class, evaluated independently per team. So when Oklahoma sits at 0.87 and Stanford at 0.83, the interpretation is that Oklahoma is a closer match to the champion profile and therefore the model judges it slightly more likely to win, not that both teams literally have an 80%+ chance of winning the same championship. The team with the highest score is taken as the predicted champion.

Although the decision boundary in feature space is linear, the mapping from a team’s features to its predicted score is nonlinear due to the sigmoid. Logistic regression is a natural fit for this problem because it produces interpretable coefficients we can inspect, outputs scores that allow teams to be compared on a continuous scale, and is well-behaved on small datasets compared to more flexible classifiers.

The training data spans 2013–2023, excluding 2020, for a total of 10 seasons, 155 team-season observations, and 10 winning team-seasons. The test set spans 2024 and 2025, adding 2 seasons, 30 observations, and 2 winning team-seasons. The combined training and test set therefore contains 12 winning team-seasons in 185 total observations, an imbalance of roughly 14 to 1. This is severe enough that an untreated logistic regression would simply learn to predict “not champion” for every team. We addressed this with scikit-learn’s `class_weight='balanced'` option, which up-weights the loss contribution of the minority class proportionally so the model has an incentive to identify champions correctly rather than ignore them.

We selected regularization strength $C = 0.1$ after evaluating $C \in \{1.0, 0.1, 0.01\}$. At $C = 1.0$, regularization was weak enough that individual feature weights grew large and the model showed clear signs of overfitting, including outsized weights on noisy features such as `std_vt` before it was

removed. At $C = 0.01$, regularization was strong enough that no feature weight could grow large enough to meaningfully separate strong teams from weak ones, and predicted probabilities clustered too narrowly to discriminate. $C = 0.1$ produced reasonable feature weights and reasonable separation between strong and weak teams.

We also briefly evaluated a Support Vector Machine (SVM) with a radial basis function (RBF) kernel as a comparison classifier [2]. Without tuning, the SVM produced clearly implausible rankings, including lower-tier programs such as Air Force and Army above established contenders. We did not pursue further tuning; the result was off badly enough that we treated it as a comparison case rather than a model worth investing in.

For prospective prediction (predicting the 2026 champion before the 2026 NCAA Championship occurred), we retrained the final model on all 12 seasons of available labeled data and applied it to the 2026 features.

III. EVALUATION

A. Initial Model Performance

The initial model used the simpler aggregate features described in Section II-B (per-apparatus mean, standard deviation, maximum, and slope, plus season-level totals and number of meets). Trained on 2013–2023 (excluding 2020), it correctly identified the champion in both held-out test seasons: Stanford in 2024 and Michigan in 2025. We then applied it to the 2026 features for prospective prediction.

B. Diagnosing the Suspicious 2026 Result

The first 2026 prediction looked wrong. Stanford was given the top probability by a wide margin despite trailing in the regular season rankings, Nebraska was placed third above Michigan, and a few other rankings were inconsistent with what the eye test suggested. We ran a diagnostic to see which features were driving these predictions. Stanford had competed in only six meets through the cutoff, fewer than any other contender, while still being heavily favored by the largest weight in the model, `max_pb`. Nebraska’s third-place ranking was traced to its vault standard deviation being unusually high relative to the field that season, which the model was multiplying by a large positive weight and inflating Nebraska’s predicted score. None of this matched what we knew about these teams.

We treated this as evidence of overfitting and ran several experiments to address it. First, we extended the training data backwards to include 2013–2015, which had previously been excluded, adding three additional championship seasons (two Michigan, one Oklahoma) to the positive examples. A direct comparison of feature weights before and after this extension confirmed instability: several weights shifted substantially, including some that flipped sign entirely. Second, we tested dropping all standard deviation features as a group. This broke the model’s test set predictions, getting both 2024 and 2025 wrong. We restored the standard deviation features and instead removed only `std_vt`, which appeared to be the specific

TABLE I
NAIVE BASELINE COMPARISON: CHAMPION IDENTIFICATION ACCURACY

Baseline (predict the team with highest...)	Accuracy
Mean total team score	7/12
NQA total (sum across all 6 apparatus)	7/12
Maximum single PB score (max_pb)	10/12
Average of top 4 PB scores	10/12
NQA parallel bars (NQA pb)	12/12

source of the noise problem. Third, we swept the regularization parameter C , which produced a substantial improvement in stability and discrimination at $C = 0.1$.

After these changes the model still correctly identified the test-set champions and produced more reasonable 2026 rankings, but `max_pb` remained one of the largest positive weights regardless of the variant.

C. Naive Baselines and the Discovery of NQA Parallel Bars

The persistence of `max_pb` as a top weight prompted us to investigate parallel bars more directly. We built a series of naive baselines that use no machine learning, predicting the champion as whichever team leads on a given statistic in a given season. The first two are general baselines for context, the next three are parallel-bars-specific.

Table I shows the result. The general baselines correctly identify the champion in 7 of the 12 labeled seasons in our dataset. Both PB-specific simple statistics jump to 10/12. The NCAA’s official NQA formula on parallel bars, computed by dropping the top regular season score, averaging the next three, and averaging that with the conference championship score, identifies the champion in all 12 seasons. Including 2026 (where Stanford led NQA parallel bars and won the championship), the streak extends to 13/13. The general baselines also include 2026 (where Oklahoma led both mean total and overall NQA but Stanford won), making them 7/13.

The full list of NQA parallel bars leaders by season is shown in Table II. The pattern holds across three different

TABLE II
NQA PARALLEL BARS LEADER VS. CHAMPION, 2013–2026

Season	NQA PB Leader	Champion
2013	Michigan	Michigan
2014	Michigan	Michigan
2015	Oklahoma	Oklahoma
2016	Oklahoma	Oklahoma
2017	Oklahoma	Oklahoma
2018	Oklahoma	Oklahoma
2019	Stanford	Stanford
2021	Stanford	Stanford
2022	Stanford	Stanford
2023	Stanford	Stanford
2024	Stanford	Stanford
2025	Michigan	Michigan
2026	Stanford	Stanford
Accuracy		13/13 (100%)

championship dynasties (Michigan, Oklahoma, Stanford) and across multiple rule change eras. This is the only single statistic, of any we examined, that has matched the champion every year. We discuss what to make of this in Section IV-A.

D. Adding NQA Features to the Final Model

Given the strength of NQA parallel bars as a single-statistic indicator, we added the full set of NQA per-apparatus features to the model, along with the conference championship total as its own feature. This was the final iteration of the model produced before the 2026 NCAA Championship. It correctly identifies all training and test season champions, the same as the previous iteration; what changed was the 2026 prediction. With NQA features included, Oklahoma’s predicted score (0.870) edged above Stanford’s (0.827), reversing the ordering produced by all earlier iterations of the model. Stanford led raw NQA parallel bars by a small margin (54.950 to Oklahoma’s 54.775), but the model’s continuous-valued treatment of NQA found Oklahoma’s overall profile across all six apparatus to be slightly stronger. We discuss the broader implication of this in Section IV-B.

E. Final 2026 Prediction

TABLE III
2026 NCAA CHAMPIONSHIP: PREDICTED VS. ACTUAL

Team	Predicted Score	Actual Finish
Oklahoma	0.870	2nd
Stanford	0.827	1st (Champion)
Michigan	0.688	3rd
Nebraska	0.255	4th

Table III shows the predictions of the final model. The top four predicted scores correspond exactly to the four actual top finishers, although the model’s single highest score went to Oklahoma rather than Stanford. The Oklahoma–Stanford gap was small (0.870 versus 0.827).

For context on how unexpected Stanford’s win was, the NCAA Coaches Poll released April 6, 2026 placed Oklahoma first with 15 first-place votes and Stanford second with 2. Both the model and the coaches favored Oklahoma, but the model’s near-tie between the two teams was closer to the eventual outcome than the lopsided distribution of first-place votes.

F. Feature Weights

Table IV shows the largest positive and negative feature weights from the final model. NQA parallel bars carries the largest positive weight, consistent with the discovery described in Section III-C. We present the weights descriptively rather than as causal claims; the discussion of weight stability in Section IV-C explains why.

G. Week-by-Week Analysis

To address the secondary question of how early in the season a reliable prediction is possible, we retrained the model at

TABLE IV
TOP FEATURE WEIGHTS, FINAL MODEL

Feature	Weight
nqa_pb	+0.580
max_pb	+0.523
conf_champ_total	+0.464
max_vt	+0.384
max_fx	+0.366
max_sr	+0.341
nqa_sr	+0.336
max_total	+0.325
std_hb	+0.319
slope_ph	+0.298
mean_total	+0.270
nqa_ph	+0.242
std_sr	-0.148
n_meets	-0.273

Apparatus codes: fx = floor exercise, ph = pommel horse, sr = still rings, vt = vault, pb = parallel bars, hb = high bar. Prefixes: nqa = National Qualifying Average, max = season maximum, std = standard deviation, slope = score trajectory.

seven cutoff dates from February 1 to April 8 using only features that could be computed from data available at each cutoff. Because the NQA features and conference championship total all depend on the conference championship having occurred, this analysis uses both a reduced and a modified feature set: the conference total is dropped, and the NQA features are replaced by a 4-score average of the team’s top apparatus scores from however many meets they have completed by the cutoff. Figure 1 shows the result for the top contenders in each season.

Across all three test seasons, the weekly scores fluctuate substantially throughout the regular season, including stretches where clearly weaker teams are ranked above contenders, and only stabilize at the final April 8 cutoff once conference championship data is incorporated into the aggregate features. The model is reliably useful only after conference championships have been competed. We discuss possible explanations in Section IV-E.

IV. DISCUSSION

A. Why It Is Surprising

A team total in NCAA Men’s Gymnastics is the sum of routines across six apparatus, so it is surprising to see a single event carry the influence parallel bars does, with no obvious explanation why. It may be that championship-caliber programs share some structural quality (depth on the event, recruiting focus, or coaching emphasis) that happens to manifest as NQA parallel bars leadership. It may also be coincidence. With only 13 data points, any single perfect indicator could plausibly be a fluke. What we can say is that this is the only single statistic, of any we examined, that has matched the champion every year, and the pattern is striking enough to document even without a causal mechanism.

Champion Probability Throughout the Season

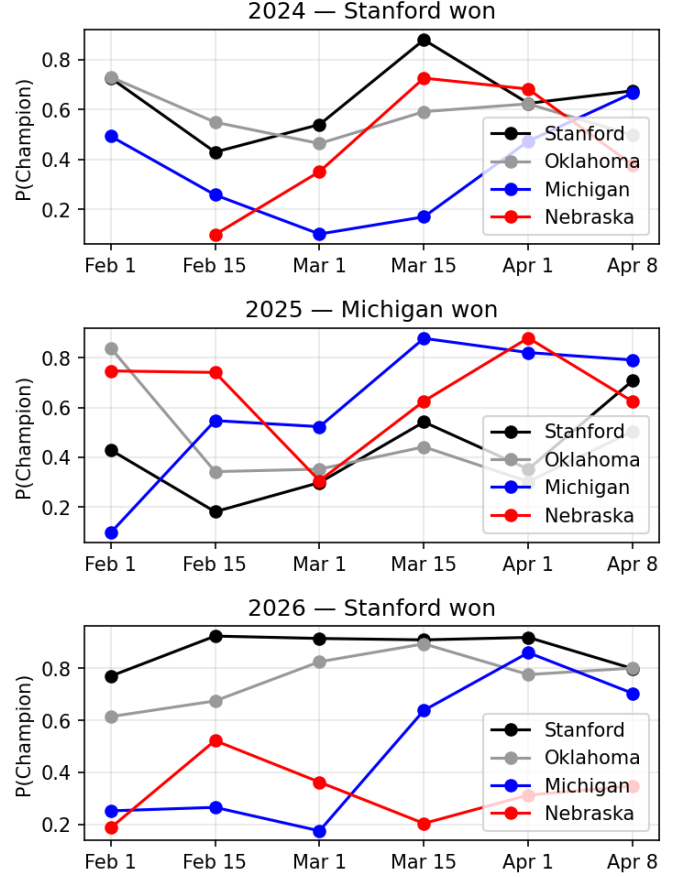


Fig. 1. Predicted champion score at seven cutoffs across the season for the top contenders in 2024, 2025, and 2026. The weekly model uses a reduced and modified feature set that excludes NQA and conference championship total, substituting a 4-score average of the team’s best apparatus scores in place of NQA where enough meets have been competed.

B. Binary Ranking versus Continuous Feature

The 13/13 NQA parallel bars finding is a ranking observation. It says that in every season examined, the team ranked first on NQA parallel bars also won the championship. This is not the same statement as “high NQA parallel bars values predict championship wins.” When NQA parallel bars enters the logistic regression, it enters as a continuous z-scored value, which preserves magnitudes and comparative gaps but discards the binary leader-versus-not signal that the historical pattern actually describes.

The 2026 prediction illustrates the gap. Stanford led NQA parallel bars at 54.950; Oklahoma was at 54.775. As a ranking, Stanford was first. As a continuous feature, Stanford’s value was only marginally higher than Oklahoma’s, and Oklahoma’s strength across the other apparatus was enough to push its predicted probability above Stanford’s. An ablation in which we removed conference championship total from the feature set left this ordering effectively unchanged (Oklahoma 87%, Stanford 83%), confirming that the continuous NQA features

themselves drive the difference rather than conference total.

We considered adding an explicit binary `is_nqa_pb_leader` indicator, but rejected it. With only 12 positive training examples, encoding a feature that hardcodes the exact pattern we are trying to validate would be overfitting in the most literal sense. The methodological point stands without it: ranking-based domain heuristics and value-based statistical features encode different information, and substituting one for the other can produce different answers.

NQA parallel bars may have been the single strongest indicator of Stanford’s 2026 win, but when folded into the model alongside the rest of the NQA features it was also the factor that pushed Oklahoma’s predicted score slightly above Stanford’s.

C. Weight Instability and Small Sample Size

The weights learned by the model are not stable in the way that one would want for causal interpretation. Adding just two seasons (2024–2025) to the training data caused several feature weights to shift substantially, including some that flipped sign entirely. With 27 features and 12 positive examples, the model has more degrees of freedom than data points, and the specific weights it lands on reflect details of the training set rather than stable underlying relationships.

This means our predictive accuracy should be evaluated separately from our feature interpretations. The model can be right about the champion while still being unreliable about why. Any claims we draw from feature weights (including the framing of the NQA parallel bars finding as something the model surfaced) must be read with this caveat in mind.

D. Stanford in the Training Data

Stanford won 5 of the 12 training-set championships, or 42%. Even with strong regularization, the model tends to favor Stanford-like feature profiles because Stanford-like profiles are over-represented among positive training examples. We made several attempts to reduce this bias, including extending the dataset backwards to add Michigan and Oklahoma championship seasons that predated our original window. The model’s tendency persisted. We cannot fully remove this bias without more diverse champion data; we can only disclose it. The near-parity of Stanford and Oklahoma in the 2026 prediction is consistent with genuine pre-meet uncertainty, but we acknowledge that some of the model’s confidence in Stanford-shaped teams reflects historical accident rather than a stable property of championship gymnastics.

E. Reliability Limits of Early-Season Predictions

The week-by-week analysis reveals that our model is unreliable until conference championships have been competed. Two explanations are plausible. The first is that the weekly model is genuinely operating on a less informative feature set: the features that change between the weekly and full models (NQA per apparatus and conference championship total) are the ones the full model weights most heavily. The

second is that something is implemented incorrectly in the weekly pipeline; some of the rankings, such as Nebraska being placed above Michigan in 2025 for most of the season, seem implausible even accounting for the modified feature set. A more careful audit of the weekly implementation, which we did not have time to perform, would help distinguish the two. Either way, the practical implication is that the model cannot reliably predict the champion until the conference championship round of competition has occurred. Whether this reflects a genuine property of the sport or a limitation of our implementation is left for future work.

F. Limits of Our Domain Knowledge

Several of the feature engineering decisions in this project were motivated by gymnastics knowledge: dropping `std_vt` after observing model behavior that did not match what we knew about the sport, using the NCAA’s NQA formula explicitly, and z-scoring within season to handle annual rule changes. Domain knowledge shaped these decisions in ways a purely statistical approach might not have arrived at directly.

Our domain knowledge also has limits. We did not have lineup-level data, so we could not actually verify our hypotheses about why `std_vt` behaved badly. We did not have D-score (difficulty) versus E-score (execution) decomposition, so our score trajectory feature cannot say whether teams are improving in execution, increasing difficulty, or both. Several of the interpretations we offer above are best read as plausible explanations rather than verified findings.

G. External Recognition

The NQA parallel bars pattern was independently noted by analysts covering men’s gymnastics directly following the 2026 championship [4]. We mention this only to indicate that the pattern is recognized within the gymnastics community, not as evidence of its predictive validity.

H. Practical Implications

The patterns surfaced here may be of interest to analysts and fans tracking championship-caliber teams across a season, in particular the parallel bars metric as a simple single-statistic indicator. We are cautious about claiming actionable implications for coaching, recruiting, or in-season strategy: a 13-season sample is too small to support prescriptive claims, and several of our findings could plausibly be coincidence. The methodological lessons of the project, including within-season normalization to handle rule changes and the distinction between ranking-based heuristics and continuous-valued statistical features, generalize beyond gymnastics to any small-sample sports prediction problem with shifting evaluation criteria.

V. CONCLUSION

We trained a logistic regression classifier to predict the NCAA Men’s Gymnastics National Champion from regular season data. The model correctly identified the champion in both held-out test seasons (Stanford in 2024 and Michigan in

2025). For the 2026 prospective prediction, the four highest probabilities corresponded to the four actual top finishers, though the model’s single highest probability went to Oklahoma rather than the eventual champion, Stanford. While investigating why parallel bars features were so persistently weighted by the model, we surfaced a striking historical pattern: the team leading NQA parallel bars has matched the champion in every season from 2013 to 2026. We are cautious about reading this as a causal result given the small sample size, but it is the only single statistic we examined that has matched the champion every year.

The project achieved its primary goal of predicting the champion with reasonable accuracy from regular season data. Its secondary goal, identifying how early in the season a reliable prediction is possible, was less successful: our model is reliably useful only after conference championships have been competed, and we cannot rule out that this reflects a flaw in our weekly pipeline rather than a genuine property of the sport.

The strongest takeaways from the project are methodological. Domain knowledge informed several engineering decisions (within-season normalization to handle annual rule changes, the use of the NCAA’s NQA formula explicitly, and the removal of `std_vt`) that a purely statistical approach might not have arrived at directly. The distinction between ranking-based heuristics and continuous-valued statistical features turned out to be more than a methodological detail: the same NQA parallel bars metric that perfectly identified champions as a binary leader indicator did not preserve that pattern when used as a continuous regression feature, and was the specific change that flipped the model’s top 2026 prediction from Stanford to Oklahoma.

Several extensions are natural. Athlete-level data would allow lineup strategy to be modeled directly rather than approximated through team-level features. The D and E score components are not currently exposed by the API; obtaining them would allow the strategic suppression of difficulty during the regular season to be modeled more directly. As additional seasons accumulate, the dataset itself will grow, which is the most reliable way to address the small sample size and Stanford-bias limitations described in Section IV.

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REFERENCES

- [1] Road to Nationals, “NCAA Men’s Gymnastics Results and Rankings.” [Online]. Available: <https://www.roadtonationals.com>
- [2] L. Gaw, “CSCI 416/516: Machine Learning, Spring 2026,” William & Mary. [Online]. Available: <https://lindagaw.github.io/courses/CSCI416-Spring26/CSCI416.html>
- [3] Anthropic, “Claude (claude-sonnet-4-6 and claude-opus-4-7),” 2026. [Online]. Available: <https://www.anthropic.com>
- [4] [Neutral Deductions], “Post on NQA parallel bars as NCAA Men’s Gymnastics championship indicator,” Instagram, 2026. [Online]. Available: <https://www.instagram.com/p/DXUQNA7EeIT/>